EX-1 data exploration using R on the dataset

```
data <- mtcars

# Basic info
head(data)
tail(data)
dim(data)
str(data)
summary(data)

# Missing values
colSums(is.na(data))

# Basic statistics
stats <- data.frame(
Mean = sapply(data, mean),
Median = sapply(data, median),
SD = sapply(data, sd)
)
print(stats)

# Plots
hist(data$mpg)
boxplot(data$mpg)
plot(data$mpg, data$hp)

# Correlation Matrix
corr_matrix <- cor(data)
print(corr_matrix)

# Heatmap
heatmap(corr_matrix)
```

EX-2 preprocessing using R on the dataset

```
data <- mtcars

# Info
str(data); summary(data)

# Missing values
colSums(is.na(data))
data[is.na(data)] <- mean(data, na.rm=TRUE)
```

```
# Convert to factor
data$cyl <- as.factor(data$cyl)

# Remove column
data$carb <- NULL

# Normalize numeric columns
data[sapply(data,is.numeric)] <- scale(data[sapply(data,is.numeric)])

# Train-Test split
set.seed(123)
i <- sample(1:nrow(data), 0.7*nrow(data))
train <- data[i,]
test <- data[-i,]

head(train); head(test)
```

EX-3 classification rule process on the dataset using SVM algorithm

```
library(e1071)
library(caret)

# Load data
data(iris)
df <- iris

# Train-test split
set.seed(123)
i <- sample(1:nrow(df), 0.8*nrow(df))
train <- df[i,]
test <- df[-i,]

# Train SVM model
model <- svm(Species ~ ., data=train, kernel="radial")

# Prediction
pred <- predict(model, test)

# Confusion Matrix
confusionMatrix(pred, test$Species)
```

```
ggplot(iris,aes(Petal.Length,Petal.Width,color=Species))+
geom_point(size=3)+theme_minimal()
```

EX-4 classification rule process on the dataset using Random forest algorithm

```
library(randomForest)
library(caret)

# Load data
data(iris)

# Train-test split
set.seed(123)
i <- createDataPartition(iris$Species, p=0.7, list=FALSE)
train <- iris[i,]
test  <- iris[-i,]

# Random Forest model
model <- randomForest(Species ~ ., data=train)

# Prediction
pred <- predict(model, test)

# Evaluation
confusionMatrix(pred, test$Species)

# Importance
varImpPlot(model)
```

EX-5 classification rule process on the dataset using Decision tree algorithm

```
library(rpart)
library(rpart.plot)

# Load data
data(iris)

# Train-test split
set.seed(123)
i <- sample(1:nrow(iris), 0.7*nrow(iris))
```

```
train <- iris[i,]
test  <- iris[-i,]

# Train model
model <- rpart(Species ~ ., data=train, method="class")

# Plot tree
rpart.plot(model)

# Prediction
pred <- predict(model, test, type="class")

# Evaluation
table(pred, test$Species)
mean(pred == test$Species)
```

EX-6 clustering rule process on the dataset using a simple k-means algorithm

```
# Load data
data(iris)
x <- iris[,1:4]

# K-means clustering
set.seed(123)
model <- kmeans(x, centers=3)

# Results
model$size
model$centers
table(model$cluster, iris$Species)

# Plot clusters
plot(x$Sepal.Length, x$Sepal.Width, col=model$cluster, pch=19)
points(model$centers[,1:2], pch=8, cex=2)
```

EX-7 Association rule process on the dataset using the Apriori algorithm

```
library(arules)

# Transaction data
transactions_data <- list(
```

```
  c("Milk","Bread","Butter"),
  c("Bread","Butter"),
  c("Milk","Bread"),
  c("Milk","Butter"),
  c("Bread","Butter","Jam"),
  c("Milk","Bread","Butter","Jam")
)

# Convert to transactions
transactions <- as(transactions_data, "transactions")
inspect(transactions)

# Apriori algorithm
rules <- apriori(transactions, parameter=list(supp=0.3, conf=0.6))
inspect(rules)

# Sort rules
rules_sorted <- sort(rules, by="confidence", decreasing=TRUE)
inspect(rules_sorted)

# Strong rules
strong_rules <- subset(rules_sorted, lift > 1)
inspect(strong_rules)
```

EX-7 Linear Regression using R on the dataset.

```
data(mtcars)

# Scatter plot
plot(mtcars$wt, mtcars$mpg, pch=19)

# Linear regression model
model <- lm(mpg ~ wt, data=mtcars)
summary(model)

# Regression line
abline(model, col="red")

# Prediction
predict(model, data.frame(wt=3))
```

EX-8 Text Mining using R on the dataset

```
library(tm); library(wordcloud)

text <- c("Text mining in data science",
        "R used for text mining",
        "Text mining extracts information")

corpus <- Corpus(VectorSource(text))
dtm <- as.matrix(DocumentTermMatrix(corpus))

freq <- sort(colSums(dtm), decreasing=TRUE)
wordcloud(names(freq), freq)
```

EX-10 Dimensionality reduction using PCA

```
data(iris)

# Numeric features
features <- iris[,1:4]

# Standardize
scaled_features <- scale(features)

# PCA
pca_model <- prcomp(scaled_features, center=TRUE, scale.=TRUE)

# Summary, Eigenvalues, Loadings
summary(pca_model)
pca_model$sdev^2
pca_model$rotation

# PCA Scores
reduced_data <- pca_model$x
head(reduced_data)

# Scree Plot
plot(pca_model, type="l")

# PCA Visualization
plot(reduced_data[,1], reduced_data[,2],
    col=as.numeric(iris$Species), pch=19,
    xlab="PC1", ylab="PC2")
```

```
legend("topright", legend=levels(iris$Species), col=1:3, pch=19)
```